Hamilton's equations permit us to calculate its position and momentum at all later times. They define a flow through the phase space.

Given this, one might guess that the quantum state of a particle would be spanned by a basis of states labeled by position and momentum:

$$|x, p\rangle.$$

The wave function would then be a function of both variables:

$$\psi(x, p) = \langle x, p|\Psi\rangle.$$

However, this is incorrect. We've already seen that things that would be simultaneously knowable in classical physics may not be in quantum mechanics. Different components of a spin, say σ_z and σ_x, are an example. One cannot know both components simultaneously; therefore, one does not have states in which both components are specified. The same is true for x and p: specifying both values is too much. Whether we're talking about spins (σ_z, σ_x) or positions and momenta (x, p), the incompatibility is ultimately an experimental fact.

What then can we know about the particle on the x axis, if not x and p? The answer is x **or** p; for according to the mathematics of position and momentum operators, the two do not commute. But I emphasize that this is not something you could have predicted in advance; it is the distillation of many decades of experimental observations.

If the position of a particle is an observable, there must be a Hermitian operator associated with it. The obvious candidate is the operator **X**. The first step in understanding this fundamental connection between the intuitive concept of position and the mathematical operator **X** is to work out the eigenvectors and eigenvalues of **X**. The eigenvalues are the possible values of position that can be observed, and the eigenvectors represent the states of definite position.

8.2.1 The Eigenvalues and Eigenvectors of Position